

Continuous Assessment Test – II

Winter Semester -2023-24

Programme Name & Branch : B.Tech ECE

Course Name & Code : BECE204L- Microprocessors and Microcontrollers

Slot: B1

Answer all the questions

Exam Duration: 50 Minutes

S.No	Question	Marks						
1	Assume a switch is connected to port P2.0, LED is connected to P1.1 and buzzer is connected to P2.2 of 8051 microcontroller. Write an assembly program for 8051 microcontroller to perform the following operations <table><tr><th>Input Switch (P2.0)</th><th>Output</th></tr><tr><td>0</td><td>Buzzer is ON for 15ms</td></tr><tr><td>1</td><td>LED is ON for 20ms</td></tr></table>	Input Switch (P2.0)	Output	0	Buzzer is ON for 15ms	1	LED is ON for 20ms	10
Input Switch (P2.0)	Output							
0	Buzzer is ON for 15ms							
1	LED is ON for 20ms							
2	Pin P1.0 of 8051 is connected to a switch. If the switch is closed (logic 0), then write an 8051 assembly language program to transmit "CLOSED" serially at the baud rate of 19200. If the switch is open (logic 1), then serially transmit "OPEN" at the baud rate of 9600. Assume "CLOSED" is stored in ROM location 200H onwards and "OPEN" in 300H location onwards.	10						
3	Write an assembly program using interrupt of 8051 microcontroller to simultaneously create 7 KHz and 500 Hz square waves on port pin P1.7 and P1.6.	10						
4.	8051 microcontroller is connected to water level sensor through ADC0808 which gives 5V when the water tank is 100% full. Develop an assembly program with suitable interfacing diagram to interface ADC0808 with 8051 to compare the water level stored in accumulator(A) with threshold value "E6H". If A>E6H send a message 'HIGH LEVEL' through serial port at a baud rate of 2400. Assume 'HIGH LEVEL' is stored in ROM location 500H onwards.	10						



5. Write an 8051 assembly program to interface 16x2 LCD display with suitable diagram. Assume the string "HELLO" stored in ROM location from 300H onwards. A switch is connected to port pin P2.7. If the switch input is 1, display the string in line 1 otherwise display the string in reverse order in line 2 as shown below.

If switch P2.7=1 display

								H	E	L	L	O				

If switch P2.7=0 display

								O	L	L	E	H				

10

School of Computer Science Engineering

Winter Semester 2023-2024

Continuous Assessment Test – II

Program Name & Branch: BTECH CSE (BCB, BCE, BCI, BCT, BDS, BKT)

Course Name & code: BECE204L Microprocessors and Microcontrollers

Slot: FI+TFI

Exam Duration: 90 Min.

Maximum Marks: 50

General instruction(s):

Answer all the questions

S. No	Question	Marks
1. A	Write the physical address of memory which is interfaced with 8086. The logical address is 1000:0200 H. Ans: Physical Address = (10H X 1000H) + 0200H = 10200H	2
B	Write the status of flag register of 8086 after executing the following program: MOV AX, 00FFH MOV BX, 00F2H ADD AX, BX Ans: CF = 0, ZF = 0, SF = 0, OF = 0, PF = 0, AF = 1, IF = 1, DF = 0.	3
C	Identify the status of the Accumulator after executing each line of the following program: MOV AL, 23H // AL = 23H 0010 0011 MOV BL, 45H // AL = 23H 0010 0011 ADD AL, BL // AL = 68H 0110 1000 XOR AL, BL // AL = 2DH 0110 1000 XOR 0100 0101 = 0010 1101 = 2DH SUB AL, 3H // 2AH MOV BL, AL // 2AH	5

2	Develop an 8086 ALP for $(A + B)^2$. Assume A and B values are 0 – 9 only. Ans: Assume (i) Data A is stored logical address at 3000H, Data B is stored at 3001H, (ii) A and B values are 5 and 6 respectively. MOV [3000H], 05H MOV [3001H], 06H MOV AL, [3000H] ADD AL, [3001H] MUL AL HLT ORG 0000H LJMP MAIN ORG 0023H LJMP SERIAL ORG 0040H	10
3	With the help of interrupt programming, develop an ALP of 8051 microcontrollers to perform the following tasks; continuously get numeric data from P0 and send it to P1. If the data is even, serially send it through port P2 with a baud rate of 19200. Otherwise, the data will be sent through P2 with a baud rate of 9600. Ans: ORG 0000H LJMP MAIN ORG 0023H LJMP SERIAL ORG 0040H	10

```

ORG 0040H
MAIN: MOV TMOD,#20H // Mode 2 of Timer 1 (Auto reload mode)
      MOV TH1,#-3 // Baud rate
      MOV SCON,#50H // Mode 1 in Serial communication
      MOV IE, #10010000B // Enable Serial Communication Interrupt
      SETB TR1 // start the timer

      XX: JB P2.0, L5
           MOV PCON,#00H //For odd number , Baud rate = 9600, PCON = 00H
           SJMP L6
      L5: MOV PCON,#80H //For even number , Baud rate = 19200, PCON = 80H
           SJMP L6
      L6: MOV P1,P0 //copy data from Port 0 to Port1 continuously
           SJMP XX

SERIAL: CLR TI
        MOV SBUF,P2 //copy data from Port 2 to UART continuously
        RETI

      END

```

4 A football stadium is designed with a seating capacity of precisely 1000 persons. It is assumed that there is just one entry and exit gate accessible within the stadium. The entry gate is configured so that whenever a person enters a stadium, a high-to-low pulse will be generated and that will be given to the port3.4 of the 8051 microcontroller. The stepper motor regulates the opening and shutting of the entry gate. To open/close the entry gate, send a logic 1/logic 0 to

10

port0.5 of the 8051 microcontroller, which connects to the stepper motor. Develop an ALP of 8051 that counts and allows maximum up to 1000 persons and closes the gate after that irrespective of persons leaving from the exit gate. Use the mode 2 timer programming.

Ans:

```

ORG 0000H
MOV TMOD, #06H // TIMER-0 AS A COUNTER
MOV TH0, #00
SETB P0.5 // Opening of the GATE
L2: SETB TR0
L1: JB TF0, L4
    MOV A, TL0
    MOV P1, A
    CJNE R1, #03h, L1 // LOGIC FOR CHECKING THE 1000 PERSONS
    L3: CJNE A, #0E8h, L1 // LOGIC FOR CHECKING THE 1000 PERSONS
    CLR P0.5 // CLOSING OF THE GATE
    SJMP Here

L4: CLR TR0
    CLR TF0
    INC R1
    SJMP L2

Here: SJMP Here
END

```

5 Develop an ALP of 8051 for displaying reversely the nine characters string (Your Reg. No) on one line LCD display with Cursor blinking. The HEX value of command words are given below.

S. No	HEX Value	Command to LCD
1.	38H	8-bit 2-line 5x7 Dots LCD
2.	30H	8-bit 1-line 5x7 Dots LCD
3.	06H	Increment cursor
4.	04H	Decrement cursor
5.	80H	Beginning of First Row
6.	8FH	End of First Row
7.	0EH	Display on Cursor on
8.	0FH	Display on Cursor blinking
9.	01H	Clear cursor

Assume Port2 is used for Data/Command word transferring, Port1.0 is connected to RS of LCD, Port1.1 is connected to $\overline{R}/\overline{W}$ of LCD and Port1.2 is connected to Enable of LCD.

Ans: Suppose Reg. No is 22BCE0123

```

ORG 0H

```

10

R = Rank mark, CO = Course Outcome, BL = Bloom's Taxonomy Level (1 = Remember, 2 = Understand, 3 = Apply, 4 = Analyse, 5 = Evaluate, 6 = Create)
Course Outcomes

CO4-Deploy the implementation of various peripherals such as general purpose input/output, timers, serial communication, LCD, keypad and ADC with 8051 microcontroller

Q.1: Assume that two sensors are connected to P3.4 and P3.5 of the 8051 Microcontroller. Sensor connected to P3.4 is placed near entry door of the auditorium and another sensor is connected to P3.5 is placed near the exit door. The maximum capacity of the auditorium is 200. Write an ALP to find the number of people inside the auditorium and send the count to port P0. An LED connected to P1.7 will glow if the auditorium is full.

Ans-1: To monitor the number of people entering and exiting an auditorium with a maximum capacity of 200, and control an LED based on this count, we can develop an Assembly Language Program (ALP) for the 8051 microcontroller. Here's how the program works:

- P3.4 is connected to the entry sensor.
- P3.5 is connected to the exit sensor.
- The count of people inside is stored in a register, and it will be updated based on the sensor input.
- The count will be output to P0.
- An LED connected to P1.7 will turn on if the auditorium is full (count = 200).

Here's the ALP code:

```
asm

ORG 0000H          ; Origin of program

MOV P0, #00H       ; Initialize count display to 0 on P0
MOV P1, #00H       ; Clear P1 (LED control)
MOV R0, #00H       ; Initialize count register (R0) to 0

; Main program loop
MAIN:
    MOV A, P3       ; Read the current state of Port 3
    JB ACC.4, ENTRY ; Check if entry sensor (P3.4) is triggered (high)
    JB ACC.5, EXIT  ; Check if exit sensor (P3.5) is triggered (high)
    SJMP MAIN       ; No change, loop back to main

ENTRY:
    INC R0          ; Increment count for entry
    MOV A, R0       ; Check if count exceeds max capacity
    CJNE A, #200, NOT_FULL
    SETB P1.7       ; Turn on LED (P1.7) if count = 200 (full)
    SJMP DISPLAY

NOT_FULL:
    CLR P1.7        ; Turn off LED if count < 200

DISPLAY:
    MOV P0, R0      ; Update P0 with the current count

WAIT_ENTRY:
    INB P3.4, MAIN  ; Wait for entry sensor to be released before continuing
```

Q.2: Two switches (SW1 and SW2) are connected to P1.2 and P1.3 of 8051 microcontroller. Develop an ALP to monitor the status of switches and transmit the following messages with the given baud rate.

S. No.	SW1	SW2	Message to be transmitted	Baud rate
1	OFF	OFF	OFF	1200
2	OFF	ON	SWITCH2	2400
3	ON	OFF	SWITCH1	4800
4	ON	ON	ON	9600

TH1	(Decimal)	(Hex)	SMOD=0	SMOD=1
-3		FD	9600	19200
-6		FA	4800	9600
-12		F4	2400	4800
-24		E8	1200	2400

S.No	SW1	SW2	Message to be transmitted serially	Baud rate
1	OFF	OFF	OFF	1200
2	OFF	ON	SWITCH2	2400
3	ON	OFF	SWITCH1	4800
4	ON	ON	ON	9600

Ans2:

Here's an Assembly Language Program (ALP) for the 8051 microcontroller to monitor the switches connected to pins P1.2 and P1.3 and transmit the specified messages at the given baud rates using serial communication. For simplicity, I'll use Timer 1 in Mode 2 (8-bit auto-reload) for baud rate generation and a polling mechanism to check switch states.

Assumptions:

1. The crystal oscillator frequency is 11.0592 MHz.
2. The serial port uses Timer 1 for baud rate control in Mode 2.

Code:

```
ORG 0000H ; Start of program
```

MAIN:

```
MOV TMOD, #20H ; Set Timer 1 in Mode 2 (8-bit auto-reload for baud rate)
```

```
MOV SCON, #50H ; Configure serial mode 1, 8-bit UART, REN enabled
```

CHECK_SWITCHES:

```
MOV A, P1 ; Load P1 to check switch states
```

Q.3: Identify what the program does and write down the comment lines for the given program

```
ORG 0H
START:
MOV SP, #60H
MOV SCON, #50H
MOV TMOD, #20H
MOV TH1, #0FDH
SETB TR1
MAIN_LOOP:
MOV DPTR, #MESSAGE
MOV R0, #0
SEND_CHAR:
MOVC A, @DPTR
INC DPTR
JNB TI, SEND_CHAR
CLR TI
CJNE A, #00H, SEND_CHAR
SJMP MAIN_LOOP
MESSAGE: DB 'MPMC', 0;
END
```

Ans-3:

```
ORG 0H ; Set the starting address of the program to 0H

START:
    MOV SP, #60H ; Set the stack pointer to 60H
    MOV SCON, #50H ; Configure the serial control register for 8-bit UART mode
    MOV TMOD, #20H ; Set Timer 1 in Mode 2 (auto-reload mode) for baud rate generation
```



Microprocessors And Microcontrollers [BECE204L] CAT-2 C1 2023-2024

1	I. A) How the physical address is calculated and find it if segment address is 4000H and offset address is 3000H. Why are these two addresses required in 8086? B) Find the errors in the following instructions (if so). I. MOV DS,5000H II. ADD [AX], [4000H] III. ADD 0200H IV. INC 3000H V. CMP BX	5	CO2	BL2
2	Write an 8086 ALP programs to find the average of 64 numbers stored in the memory location 5000H with and without using DIV instruction. Assume that the sum does not exceed 16 bits.	10	CO2	BL4
3	Write an 8051 ALP to find (compute) the numbers which are divisible by both 3 and 7 in the range 1 to 99 and store them in memory. Interface two seven segment displays in P0 and P1 and display the stored numbers one after other.	10	CO2	BL4
4	Write an 8051 (AT89C51) ALP to generate a pulse waveform with frequency 2KHz using timer. Don't access TL0 or TL1 register in the ALP. Provide the necessary timer calculations.	10	CO4	BL4
5	Write 8051 (AT89C51) ALP for the following logic. The existing values of TCON=01H and IE = 0FFH. IF (P3.2 ==1) GENERATE WAVEFORM WITH 50% DUTY CYCLE. ELSE GENERATE WAVEFORM WITH 33% DUTY CYCLE.	10	CO4	BL4

Explore More

CAT-2:Answer Key (Slot-B2)

1. Develop an 8051 Assembly Language Program (ALP) that reads data from Port P1 and performs the following:

= If the most significant bit (P1.7) is 1, perform a bitwise AND of data read from port P1 with

the value at Port P2 and store the result at Port P3.

= If the most significant bit (P1.7) is 0, perform a bitwise OR of data read from port P1 with

the value at Port P2 and store the result at Port P3.

Solution:

IterateIterate

IterateIterateIterateIterateIterate 11 ORG 0000 H ; Start of the program

IterateIterateIterateIterateIterate 12 MAIN :

IterateIterateIterateIterateIterate 13 MOV P1 , #0 FFH ; Configure P1 as input

IterateIterateIterateIterateIterate 14 MOV P2 , #0 FFH ; Configure P2 as input

IterateIterateIterateIterateIterate 15 MOV P3 , #00 H ; Configure P3 as output

IterateIterateIterateIterateIterate 16

IterateIterateIterateIterateIterate 17 READ_LOOP :

IterateIterateIterateIterateIterate 18 MOV A , P1 ; Read data from Port P1

IterateIterateIterateIterateIterate 19 JB ACC.7 , AND_OP ; IF P1.7 = 1 , jump to AND operation

IterateIterateIterateIterateIterate 110

IterateIterateIterateIterateIterate 111 ; IF P1.7 = 0 , perform OR operation

IterateIterateIterateIterateIterate 112 ; =====

IterateIterateIterateIterateIterate 113

IterateIterateIterateIterateIterate 113 MOV A , P1 ; Load data from P1 again

IterateIterateIterateIterateIterate 114 ORL A , P2 ; Perform bitwise OR with

P2

IterateIterateIterateIterateIterate 115 MOV P3 , A ; Store result in P3

IterateIterateIterateIterateIterate 116 SJMP READ_LOOP ; Repeat loop

IterateIterateIterateIterateIterate 117

IterateIterateIterateIterateIterate 118 ; IF P1.7 = 1 , perform AND operation

IterateIterateIterateIterateIterate 119 ; =====

IterateIterateIterateIterateIterate 120

IterateIterateIterateIterateIterate 120 AND_OP :

IterateIterateIterateIterateIterate 121 MOV A , P1 ; Load data from P1 again

IterateIterateIterateIterateIterate 122 ANL A , P2 ; Perform bitwise AND with

P2

IterateIterateIterateIterateIterate 123 MOV P3 , A ; Store result in P3

IterateIterateIterateIterateIterate 124 SJMP READ_LOOP ; Repeat loop



School of Electronics Engineering (SENSE)



Fall Semester 2024-25

Continuous Assessment Test – II

SLOT: G2+TG2

Programme Name & Branch: B. Tech & CSE

Course Name & Code: Microprocessor & Microcontrollers (BEE204L)

Class Number (s): VL2024250104331, VL2024250104315, VL2024250104323, VL2024250104343, VL2024250104333, VL2024250104318, VL2024250104334, VL2024250104326, VL2024250104326

Faculty Name (s): Debashish Dash; A-D-D Dwivedi; G Ragunath, Vikas Vijayavargiya, Henridass, Abhishek Tripathi, Naushad H Laskar, Vishal Gupta

Exam Duration: 90 Min.

Maximum Marks: 50

General instruction(s):

Specify if any printed material may be permitted

Any other specific instruction

Q. No.	Question	Max Marks	CO	BL
1.	Using Timer 0 auto reload mode of the 8051-microcontroller having 12Mhz crystal frequency, write an Assembly Language Program (ALP) to generate a 2 kHz square wave on pin P1.0. Additionally, implement the following features: (i) The program should count how many times the square wave has toggled (both rising and falling edges). After 50 toggles, the frequency of the square wave should automatically double. (ii) After another 50 toggles (total of 100 toggles), the program should stop generating the square wave	10	CO4	BL4
2.	Assume that a switch (SW) is connected to pin P1.7. Write an assembly language program for 8051 to monitor the switch and perform the following: (a) If SW=0, send the message "HELLO" serially with baud rate of 9600. (b) If SW=1, send the message "GOODBYE" serially with baud rate of 4800.	10	CO4	BL5
3.	Develop an assembly language program for the following scenario: Imagine a sensor is associated with pin P3.2 (INT0) to count the number of entries inside the hall and a buzzer is connected to pin P1.5 of 8051. For sensor, you can feed the input by connecting a switch at P3.2. Normally, the switch is high, and buzzer is OFF. Sensor will count the number of incoming persons entering the hall only when it goes low and makes the buzzer ON.	10	CO4	BL5

4.	Write an assembly language program for 8086 to compute the function $y = (X^3 + X + 1)/2$. Assume the values of X are the 5 initial odd numbers. Also, store the result at 3000H onwards.	10	CO2	BL4
5.	a) The following segment and offset values are provided for an 8086 microprocessor: Code Segment: 2A3FH, Data Segment: 3B20H, Instruction Pointer: 0042H, Extra Segment: 4C10H, Stack Segment: 3000H, Base Register: 015AH, Destination Index: 00E5H, Source Index: 00D0H, Stack Pointer: 1000H Calculate the physical address for the following: MOV AX, [CS:IP] MOV AX, [DS:BX] MOV [ES:DI], AX MOVSX PUSH BX b) Find the errors in the following instructions for 8086 (if so). Please correct these. i). MOV DS, 5000H ii). ADD [AX], 4000H iii). DIV BX, CX iv). INC 4500H v). CMP BX	5+5	CO2	BL4

SCHOOL OF ELECTRONICS ENGINEERING (SENSE)

Continuous Assessment Test - II
Winter Semester -2022-23

Programme Name & Branch : B.Tech ECE

Course Name & Code : BECE204L- Microprocessors and Microcontrollers

Slot: C1

Answer all the questions

Exam Duration:50 Minutes

S.No	Question	Marks
1.	Write an 8051 assembly language program to toggle P2.3 every 225 micro secs, use Timer1 for the delay creation, assume XTAL = 11.0592 MHz.	10
2.	Write an 8051 assembly language program to receive data serially, check the LSB bit of the received data, if it is 0, send the received data to Port P1, else send the received data to Port P2, use 2400 as the baudrate, assume XTAL=11.0592 MHz.	10
3.	Write an 8051 assembly language program to get the data from Port P0 and send it to Port P1 continuously, while the interrupts will do the following, assume XTAL = 11.0592 MHz (a) Keep transmitting the data 'VIT' continuously through the serial COM port with 2400 baudrate. (b) Whenever there is a falling edge at P3.3 (INT1 pin), turn on the buzzer at P1.4 for some time.	10
4.	Write an 8051 assembly language program to use Timer 0 as a counter in mode 2, show the state of the counter in Port P2, whenever the counter rolls over increment the value in Port P1. Assume XTAL = 11.0592 MHz	10
5.	Assume a switch is connected to P2.4, write an 8051 assembly program to monitor the status of the switch, if the switch is ON (logic 1), then generate a square wave of 2 KHz at P2.1, else generate a square wave of 5 KHz at P2.7, use XTAL = 12 MHz.	10

1. Write an 8051 assembly language program to toggle P2.3 every 225 micro secs, use timer 1 for the delay creation, assume XTAL = 11.0592 MHz

$\therefore$ Time delay = 225 μ sec

Number of Machine cycles = $\frac{225 \mu\text{sec}}{1.09 \mu\text{sec}}$

CONTINUOUS ASSESSMENT TEST - II
FALL SEMESTER 2024-2025

SLOT: G1+TG1

Programme Name & Branch
Course Code and Course Name

: R. Tech & CSE
: Microprocessor & Microcontrollers (BECE204L)

Faculty Name(s)
Ghosh, Vikas Vijayavargiya, Hendrass, Abhishek Tripathi, Naushad H Laskar, Vishal Gupta

Class Number(s)
VL2024250104321, VL2024250104315,
VL2024250104323, VL2024250104343, VL2024250104331, VL2024250104318,
VL2024250104334, VL2024250104328, VL2024250104326

Date of Examination
Exam Duration

: 19-Oct-2024
: 90 minutes

Maximum Marks: 50

General Instruction(s):

- Answer All Questions
- M - Max mark; CO - Course Outcome; BL - Blooms Taxonomy Level (1 - Remember, 2 - Understand, 3 - Apply, 4 - Analyse, 5 - Evaluate, 6 - Create)
- Assume missing data wherever applicable.

Q.No	Question	M	CO	BL
1	Write an assembly language program for the 8051 to generate a square wave of frequency 100 Hz on pin P1.0 by using timer 0 mode 2. Assume that XTAL = 11.0592MHz.	10	CO4	BL5
2	Write an 8051-assembly level program to transmit a sequence of 4 characters ('A', 'B', 'C', 'D') serially. After completing this transmission double the baud rate and then transmit another sequence of 4 characters ('E', 'F', 'G', 'H'). Toggle an output pin P1.0 every time a character is successfully transmitted. (Crystal frequency of microcontroller is 11.0592 MHz)	10	CO4	BL5
3	Write an 8051-assembly program using Timer 0 to generate an interrupt every 120 micro second and External Interrupt 1 (INT1) to trigger on a falling edge. Upon each Timer 0 interrupt, increment a counter and display the counter value on Port 2. When External INT1 occurs, immediately reset the counter and turn ON an LED connected to P1.7. The LED should remain ON for some time after the interrupt. To ensure proper functionality, configure the system such that Timer 0 interrupts do not interfere with the handling of External INT1, which should have the highest priority. (Crystal frequency of microcontroller is 11.0592 MHz)	10	CO4	BL5

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SCHOOL OF ELECTRONICS ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
FALL SEMESTER 2024-2025

SLOT: G1+TG1

4	(a) In the following program identify the errors (if any) and re-write the correct instruction. If no error found, write "correct" against the instruction. (i) ADD AX, 1234H (ii) SUB BL, AX (iii) MOV DS, 5060H (iv) PUSH [BP+04H] (v) CMP CX, 5000D	5	CO2	BL4
5	(b) If CS = 3500H and DS = 4000H. Find if there is any overlap between these two segments. If so, what is the size of the overlap?	5		
6	Write an 8086-assembly language program to count the number of 1's (set bits) in an 8-bit binary number stored at memory location 2000H. Store the result (count of 1's) at memory location 3000H.	10	CO2	BL4



VIT
Vellore Institute of Technology
Vellore, Tamil Nadu 620 015, India

REG. NO.:

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

SLOT: B1+TB1

Programme Name & Branch : B.Tech CSE
Course Code and Course Name : BECE204L- Microprocessors and Microcontrollers
Faculty Name(s) : Gerardine Immaculate Mary (Course Co-ordinator)
Class Number(s) : VL2024250504038/4040/4042/4045/4047/4049/4051/5054/
4057/4059/4061/4063/4065/4068/4071/4073/4075/4077/4078
4080/4082/4084/4086/4088/4191/4199
Date of Examination : 17-03-2025
Exam Duration : 90 minutes
Maximum Marks: 50

General instruction(s):

- Answer All Questions
- M - Max mark; CO - Course Outcome; BL - Blooms Taxonomy Level (1 - Remember, 2 - Understand, 3 - Apply, 4 - Analyse, 5 - Evaluate, 6 - Create)
- Course Outcome
CO4-Deploy the implementation of various peripherals such as general purpose input/output, timers, serial communication, LCD, keypad and ADC with 8051 microcontroller
- Comment the program instructions, explain the format of Special Function Registers (SFRs) used, and provide any necessary calculations.

Q. No	Question	M	CO	BL
1.	Develop an 8051 Assembly Language Program (ALP) for the following scenario. The port P1 receives the data from the user. Based on the nature of the received data do the following: - If the received data at P1 is EVEN then send it to P2 by inverting bits that are in the even position. - If the received data at P1 is ODD then send it to P3 by inverting bits that are in the odd position.	10	4	3
2.	Develop an 8051 ALP to interface a capacitive touch sensor with Port P3.4 and use Timer 0 to detect touch events. When the sensor is touched, increment the count on Port P1 and display the touch count on Port P2 every 1 second using Timer 1.	10	4	3
3.	Develop an 8051 ALP to receive data via the serial port at a baud rate of 9600. The received data should be buffered in Port P1, and the program should send an acknowledgment message ("Data Received") back to the sender using interrupt.	10	4	3
4.	Develop an 8051 ALP to interface 16x2 LCD display with suitable diagram. Assume the string "VIT" is stored in ROM location from 200H onwards and the string "Vellore" is stored in ROM location from 300H onwards. A switch is connected to port pin P0.4. If the switch input is 1, display the string "VIT" in line 1 of the LCD display otherwise display the string "Vellore" in line 2 of the LCD display.	10	4	3
5.	An 8051 microcontroller is connected to linear temperature sensor through ADC0804 which gives 5V when the temperature of the water in boiler at industry is 100°C. Develop an ALP with suitable interfacing diagram to interface ADC0804 with 8051 and compare the value of temperature stored in Accumulator (A) with threshold value T=50°C. If A>50°C then turn ON buzzer connected to port P1.3 to alarm for 60 ms using timer 0.	10	4	3

Q1: Develop an 8051 Assembly Language Program (ALP) for the following scenario. The port P1 receives the data from the user. Based on the nature of the received data do the following:

- If the received data at P1 is EVEN then send it to P2 by inverting bits that are in the even position.
- If the received data at P1 is ODD then send it to P3 by inverting bits that are in the odd position.

OR_G 0000H
MOV P1, #0FFH } to make a Port 7H Port
MOV A, P1 } Putting 7H data in a register
MOV R0, A }
... add no. the LSB



VIT
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(Established in the University sector under Section 3 of UGC Act, 1956)

SCHOOL OF ELECTRONICS ENGINEERING (SENSE)

Continuous Assessment Test – II

Winter Semester -2023-24

Programme Name & Branch : B.Tech ECE

Course Name & Code : BECE204L- Microprocessors and Microcontrollers

Slot: B2

Answer all the questions

Exam Duration: 50 Minutes

S.No	Question	Marks
1	Write an 8051 assembly language program to generate wave of frequency 2 KHz and 40% duty cycle in P1.0 using Timer 1 for the delay creation. Assume XTAL = 20MHz.	10
2	The car parking area can accommodate a maximum of 300 cars. Assume that a sensor is connected to P3.4 of the 8051 microcontroller to sense the incoming car entering a parking area. If the count reaches 300(decimal), the microcontroller should send message "FULL" through serial port at a baud rate of 19200. Assume the message is stored in ROM location 400H onwards. Write a suitable 8051 assembly code to implement for the above scenario.	10
3	Write an assembly program using interrupt of 8051 microcontroller to simultaneously create 5 KHz and 700 Hz square waves on port pin P1.5 and P1.4.	10
4	Write an 8051 assembly code with suitable diagram to interface common anode seven segment display, which should display alphabet A to C with an interval of 0.7 seconds between each alphabet. Assume the alphabet A to C codes are stored in ROM location 200H onwards. Use timer1 to generate time delay.	10
5	8051 microcontroller is connected to linear temperature sensor through ADC0808 which gives 5V when the temperature of the water in boiler at industry is 100°C. Develop an assembly program with suitable interfacing diagram to interface ADC0808 with 8051 to compare the temperature stored in Accumulator(A) with threshold value $T=50^{\circ}\text{C}$. If $A > 50^{\circ}\text{C}$ turn ON buzzer connected to port P0.7, to alarm for 70 ms using timer 0.	10

CONTINUOUS ASSESSMENT TEST - II

FALL SEMESTER 2024-2025

Programme Name & Branch : B.Tech & ECE
Course Code and Course Name :BECE204L Microprocessors and Microcontrollers
Faculty Name(s) :Dr. Sumit Kumar Jindal & Dr. Arun Dev Dhar Dwivedi
Class Number(s) :VL2024250104199, VL2024250104201
Date of Examination :15/10/2024
Exam Duration : 90 minutes Maximum Marks: 50
General instruction(s):
Answer All Questions

M - Max mark; CO - Course Outcome; BL - Blooms Taxonomy Level (1 - Remember, 2 - Understand, 3 - Apply, 4 - Analyse, 5 - Evaluate, 6 - Create)
Course Outcomes

CO4-Deploy the implementation of various peripherals such as general purpose input/output, timers, serial communication, LCD, keypad and ADC with 8051 microcontroller

Q.1: Assume that two sensors are connected to P3.4 and P3.5 of the 8051 Microcontroller. Sensor connected to P3.4 is placed near entry door of the auditorium and another sensor is connected to P3.5 is placed near the exit door. The maximum capacity of the auditorium is 200. Write an ALP to find the number of people inside the auditorium and send the count to port P0. An LED connected to P1.7 will glow if the auditorium is full.

Ans-1: To monitor the number of people entering and exiting an auditorium with a maximum capacity of 200, and control an LED based on this count, we can develop an Assembly Language Program (ALP) for the 8051 microcontroller. Here's how the program works:

- **P3.4** is connected to the entry sensor.
- **P3.5** is connected to the exit sensor.
- The count of people inside is stored in a register, and it will be updated based on the sensor input.
- The count will be output to **P0**.
- An LED connected to **P1.7** will turn on if the auditorium is full (count = 200).

Here's the ALP code:

```
asm

ORG 0000H          ; Origin of program

MOV P0, #00H       ; Initialize count display to 0 on P0
MOV P1, #00H       ; Clear P1 (LED control)
MOV R0, #00H       ; Initialize count register (R0) to 0

; Main program loop
MAIN:
    MOV A, P3       ; Read the current state of Port 3
    JB ACC.4, ENTRY ; Check if entry sensor (P3.4) is triggered (high)
    JB ACC.5, EXIT  ; Check if exit sensor (P3.5) is triggered (high)
    SJMP MAIN       ; No change, loop back to main
```

```
ENTRY:
    INC R0          ; Increment count for entry
    MOV A, R0       ; Check if count exceeds max capacity
```

Q.2: Two switches (SW1 and SW2) are connected to P1.2 and P1.3 of 8051 microcontroller. Develop an ALP to monitor the status of switches and transmit the following messages with the given baud rate.

S. No.	SW1	SW2	Message to be transmitted	Baud rate
1	OFF	OFF	OFF	1200
2	OFF	ON	SWITCH2	2400
3	ON	OFF	SWITCH1	4800
4	ON	ON	ON	9600

TH1	(Decimal)	(Hex)	SMOD=0	SMOD=1
-3		FD	9600	19200
-6		FA	4800	9600
-12		F4	2400	4800
-24		E8	1200	2400

S.No	SW1	SW2	Message to be transmitted serially	Baud rate
1	OFF	OFF	OFF	1200
2	OFF	ON	SWITCH2	2400
3	ON	OFF	SWITCH1	4800
4	ON	ON	ON	9600

Ans2:

Here's an Assembly Language Program (ALP) for the 8051 microcontroller to monitor the switches connected to pins P1.2 and P1.3 and transmit the specified messages at the given baud rates using serial communication. For simplicity, I'll use Timer 1 in Mode 2 (8-bit auto-reload) for baud rate generation and a polling mechanism to check switch states.

Assumptions:

1. The crystal oscillator frequency is 11.0592 MHz.
2. The serial port uses Timer 1 for baud rate control in Mode 2.

Code:

ORG 0000H ; Start of program

MAIN:

MOV TMOD, #20H ; Set Timer 1 in Mode 2 (8-bit auto-reload for baud rate)

MOV SCON, #50H ; Configure serial mode 1, 8-bit UART, REN enabled

CHECK_SWITCHES:

MOV A, P1 ; Load P1 to check switch states

JB ACC.2, SW1_ON ; Check if SW1 (P1.2) is ON

JB ACC.3, SW2_ON ; Check if SW2 (P1.3) is ON

SWITCHES_OFF:

SW1 OFF SW2 OFF

Q.3: Identify what the program does and write down the comment lines for the given program

```
ORG 0H
START:
MOV SP, #60H
MOV SCON, #50H
MOV TMOD, #20H
MOV TH1, #0FDH
SETB TR1
MAIN_LOOP:
MOV DPTR, #MESSAGE
MOV R0, #0
SEND_CHAR:
MOVC A, @DPTR
INC DPTR
JNB TI, SEND_CHAR
CLR TI
CJNE A, #00H, SEND_CHAR
SJMP MAIN_LOOP
MESSAGE: DB 'MPMC', 0;
END
```

Ans-3:

```
ORG 0H ; Set the starting address of the program to 0H

START:
MOV SP, #60H ; Set the stack pointer to 60H
MOV SCON, #50H ; Configure the serial control register for 8-bit UART mode
MOV TMOD, #20H ; Set Timer 1 in Mode 2 (auto-reload mode) for baud rate generation
MOV TH1, #0FDH ; Load the timer high register TH1 with 0FDH to set the baud rate
SETB TR1 ; Start Timer 1 to begin baud rate generation

MAIN_LOOP:
MOV DPTR, #MESSAGE ; Load the data pointer register DPTR with the address of the message
MOV R0, #0 ; Initialize R0 to 0 (used as an index for characters)

SEND_CHAR:
MOVC A, @DPTR ; Move the character at the address in DPTR to accumulator A
```

Reg. No.: [REDACTED]
Name: [REDACTED]

VIT
Vellore Institute of Technology
(Approved by the Tamil Nadu State Council of Higher Education)

Continuous Assessment Test II - March 2023

Programme	B. Tech. (CSE)	Semester	WS 2022-23
Course	Microprocessors and Microcontrollers	Code	BECE 204L
Faculty	Dr. Dhireen Ku Mahapatra	Class Nbr	C11202235001403
Time	90 Minutes	Slot	G2+TG2
		Max. Marks	50

Answer ALL the questions

Q.No.	Sub. Sec.	Questions	Marks
1		Consider three 8-bit numbers X, NUM1 and NUM2 are stored in internal data RAM locations 20H, 21H and 22H respectively. Write an 8051-assembly language program to compute the following if X is equal to 0; then LSB of NUM1 (AND) LSB of NUM2 if X is equal to 1; then MSB of NUM1 (OR) MSB of NUM2 if X is equal to 2; then complement MSB of NUM1 else do nothing store the bit results in RES, where RES is MSB of 23 H locations.	10
2		Explain the types of addressing modes and type of instruction set used in the following 8051 assembly language instructions with its brief explanation ✓ MOV P0, #0F7H ✓ MOV @R1, @R0 ✓ MOV @R1, 80H ✓ MOV C, A+DPTR ✓ MOV TMOD, #0ADH ✓ MOV IE, #88H ✓ SETB TR1 ✓ XRL A, P1 ✓ ANL C, P1.0 ✓ SJMP HERE	10
3		Write an 8051-assembly language program to generate a square wave with pulse width of 1 Sec on P2.3. What value do we need to load the timer's register if we want to have a time delay of 1 Sec? Assume that XTAL = 11.0592 MHz.	10
4		Write an 8051-assembly language program to transfer "8051" serially at 9600 baud, 8-bit data, 1 stop bit, do this continuously. Assume appropriate mode of serial	10

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	communication. Discuss in detail about values to be loaded in serial communication related registers.	
5	Write an 8051-assembly language program with timer 0 to turn on an LED connected to P0.5 for one second, turn it off for 500ms repeatedly. If an external interrupt from INT 0 is provided, turn on an LED connected to P0.6 for one second and then turn it off. Assume XTAL=11.0592MHz.	10



School of Electronics Engineering

Winter Semester 2023-24 Continuous Assessment Test – II

Programme Name & Branch: B. Tech (School of Computer Science Engineering)

SLOT: C2+TC2

Course Name & Code: Microprocessors and Microcontrollers BECE204L

Exam Duration: 90 Min.

Maximum Marks: 50

Q. No.	Questions	Max Marks	CO	BL
1	a) CS = 4000H and DS = 4500H. Find if there is any overlap between these two segments. If so, what is the size of the overlap. b) Find the errors in the following instructions (if so). i. MOV CS,5000H ii. SBB [AX], [2000H] iii. MUL AL, BL iv. AL = 83H; CL = 29H; ADD AL, CL; DAA; After execution (AL)=05, carry flag (CY)=0; v. OR AX, 0098H contents of AX = 3F0FH, after execution AX = 3098H.	5 5	CO2	BL2
2	Write an 8086 ALP to find (compute) the values of the function $f(x) = x^2 + 3x + 2$ in the range $x=1$ to $x=10$ and store them in the memory location from 5000H onwards.	10	CO2	BL4
3	Write an 8051 ALP to multiply two numbers without using MUL AB instruction and assume that the product does not exceeds 8-bits. Display the digits of the product on the seven segment displays connected to P0 and P1.	10	CO4	BL4
4	Write an 8051 (AT89C51) ALP to generate a periodic waveform simultaneously as follows: i) 0.1ms ON & 0.1ms OFF ii) 0.2ms ON & 0.2ms OFF using timer. Don't access TL0/1 register in the ALP. Provide necessary timer calculations.	10	CO4	BL4
5	Write 8051 (AT89C51) ALP for the following logic. The existing values of TCON=01H and IE = 0FFH. IF (P3.3 == 1) SEND THE SEQUENCE 0,1,2,3,5,6,7,8,9 REPEATLY IN P0. ELSE SEND THE SEQUENCE 9,8,7,6,5,4,3,2,1 REPEATLY IN P1.	10	CO4	BL4

Key

Q1

a) CS = 4000H and DS = 4500H. Find if there is any overlap between these two segments. If so, what is the size of the overlap.

Code segment range = 4000H to 4FFFFH and data segment range = 45000H to 54FFFFH
Overlap size = 4FFFF - 45000 = AFFFF or 43 KB

b) Find the errors in the following instructions (if so).
i. MOV CS,5000H --- MOV CS, AX
ii. SBB [AX], [2000H] --- SBB AX, [2000H]
iii. MUL AL, BL --- MUL BL
iv. AL = 83H; --- AL = 12, CY = 1
CL = 29H;
ADD AL, CL;
DAA;

1	The contents of different registers are given below. For the given instructions, obtain the physical address and specify the addressing mode of each instruction. [4x2.5marks] AX=1000H, BX=2F00H, SI=6000H, DI=4000H, BP=5000H, SP=2000H, CS=0000H, DS=1000H, SS=3000H, IP=7000H i.) MOV AX, 5000[BX] ii.) MOV AX, 500[BX+80h] [SI] iii.) MOV AX, 50[BP] [SI+08h] iv.) MOV AX, [SI+98h]	[10]	2	K2																								
2	Ravi invited his friends to a restaurant to celebrate his birthday and they ordered food items as shown in Table 1. Help Ravi to calculate the expected total expenditure through 8086 assembly language program for this party. Table .1 <table><tr><th>Sl No</th><th>Items</th><th>Quantity</th><th>Price per unit(INR)</th></tr><tr><td>1</td><td>Chicken-65</td><td>5</td><td>45</td></tr><tr><td>2</td><td>Biryani</td><td>4</td><td>80</td></tr><tr><td>3</td><td>Ice cream</td><td>4</td><td>65</td></tr><tr><td>4</td><td>Mutton Curry</td><td>2</td><td>75</td></tr><tr><td>5</td><td>Meals</td><td>1</td><td>50</td></tr></table>	Sl No	Items	Quantity	Price per unit(INR)	1	Chicken-65	5	45	2	Biryani	4	80	3	Ice cream	4	65	4	Mutton Curry	2	75	5	Meals	1	50	[10]	2	K2
Sl No	Items	Quantity	Price per unit(INR)																									
1	Chicken-65	5	45																									
2	Biryani	4	80																									
3	Ice cream	4	65																									
4	Mutton Curry	2	75																									
5	Meals	1	50																									

	Assume all the values mentioned here are in hexadecimal. Store the Quantity in the offset location 2000H and Price per unit in the offset location 3000H. Use appropriate assembler directives.			
3	Consider 8 LEDs are interfaced with 8086 using 8255 in BSR mode. Write an 8086 assembly language program to produce a running display (one LED at a time from right to left). Assume the base address of 8255 is 8080h.	[10]	2	K3
4	Design an 8051-based bus ticketing system with the following hardware and functionality: Hardware setup: 2x2 matrix keypad: rows → P1.0, P1.1; columns → P0.0, P0.1. - Character LCD: data lines D0–D7 → P2; RS → P3.7; R/W → P3.6; E → P3.5. Operation: - Key mapping: - Key 1 → Route 1 — Fare: Rs 50 - Key 2 → Route 2 — Fare: Rs 30 - Key 3 → Route 3 — Fare: Rs 25 - Key 4 → Route 4 — Fare: Rs 40 - When a key (1–4) on the keypad is pressed the system displays the selected route on the first line of the LCD and the corresponding fare on the second line. Produce a clear interfacing diagram showing the 8051 MCU connections to the 2x2 keypad and the character LCD. Indicate the port pins used. (5 marks)	[5]	4	K4
5	Use the same configuration of Question 4 and write a complete 8051 Assembly Language Program (ALP) that: - Scans the 2x2 keypad and detects single key presses (with debounce and key-release handling). - Maps each key (1–4) to the correct route and fare strings. - Displays the route on LCD line 1 and the fare on LCD line 2.	[15]	4	K4

*****All the best *****

Course Code and Course Name : BECE201L - Microprocessors and Microcontrollers
 Faculty Name(s) : Dr. Germaine Immaculate Mary, Dr. Shanmugasundaram M, and Dr. Rajesh N
 Class Number(s) : VL2022260103664 / 3666 / 3867
 Date of Examination : 10/10/25
 Exam Duration : 90 minutes
 Maximum Marks: 50

General instruction(s):

- CO4 - Deploy the implementation of various peripherals such as general purpose input/output, timers, serial communication, LCD, keypad, and ADC with 8051 microcontroller.

Answer ALL Questions

Q. No	Question	M	CO	BL
1.	(a). Draw the format of TMOD register of 8051 and also calculate the value to be assigned to configure the following: (i). Timer 0 in mode 1 and Counter 1 in mode 2 (ii). Counter 0 in mode 2 and Timer in mode 1	5	CO4	2
	(b). Draw the format of IE register of 8051 and also calculate the value to be assigned to enable the following interrupts in an application: (i). Timer 0, external interrupt 1 and serial communication Reception (ii). Counter 1, and external interrupt 0.	5	CO4	2
2.	Develop an 8051 Assembly Language Program (ALP) to perform the following task: Assume a sensor connected to port pin P1.0 provides a HIGH (1) or LOW (0). If P1.0 is high, generate a 1 KHz wave of 25 % duty cycle on P2.0 using Timer 0; else generate a waveform of 50% duty cycle of 1 KHz on P2.1 using Timer 1.	10	CO4	3
3.	Write an 8051 ALP to add two numbers received from Port P0 and P1. If carry is generated then transfer a message "CARRY GENERATED" serially at 19200 baud rate. If no carry is generated, receive a message serially at 9600 baud rate and transfer to P2. Do it repeatedly.	10	CO4	3
4.	Assume a sensor is connected to INT0 of 8051 microcontroller. When the sensor generates momentarily high-to-low signal, transfer data from port P0 to P1 and then send a low-to-high signal at P2.1, simultaneously generating a delay of 100 microseconds at P2.2 using timer 0. Write an ALP for the above scenario.	10	CO4	3
5.	Display each character in the string "sense" with a time delay of 2 ms between each character on a 7-segment LED common cathode display. Assume port pin P2.0-P2.6 of 8051 is connected to LED segments a to g respectively.	10	CO4	3